

Skills checks cover pre-requisite material needed for the first unit of the course, limits. Answer the following questions neatly and completely. Calculators are not allowed.

1. Graph the following piecewise function: $f(x) = \begin{cases} x & \text{if } x \geq 2 \\ x^2 - 3 & \text{if } 0 \leq x < 2 \\ 5 & \text{if } x < 0 \end{cases}$

2. Graph the following piecewise function: $f(x) = \begin{cases} \frac{1}{2}x - 4 & \text{if } x \geq 2 \\ 3 & \text{if } x < 2 \end{cases}$

3. Graph the following piecewise function: $f(x) = \begin{cases} 3x-5 & \text{if } x > 4 \\ x^2 & \text{if } x \leq 4 \end{cases}$

4. Complete the following subtraction and fully simplify the result: $\frac{3}{t+3} - \frac{1}{4t+12}$

5. Complete the following addition and fully simplify the result: $\frac{x}{2x^2+2x} + \frac{3}{x+1}$

6. Complete the following subtraction and fully simplify the result: $\frac{5}{x-2} - \frac{4x+13}{x^2+x-6}$

7. Determine the interval communicated by the inequality: $|x+3| < 5$

8. Determine the interval communicated by the inequality: $3|2x-1| < 9$

9. Determine the interval communicated by the inequality: $|1-2x| < 5$

10. Rationalize the denominator of the fraction: $\frac{x-4}{\sqrt{x}-2}$

11. Rationalize the numerator of the fraction: $\frac{\sqrt{t^2+9}-3}{t^2}$

12. Rationalize the denominator of the fraction: $\frac{w-2}{\sqrt{w+2}-2}$

13. Setup and simplify the expression $\frac{f(x+h)-f(x)}{h}$ using the function: $f(x) = 5x^2 - 1$

14. Setup and simplify the expression $\frac{f(x+h)-f(x)}{h}$ using the function: $f(x) = 4 - 2x^2$

15. Setup and simplify the expression $\frac{f(x+h)-f(x)}{h}$ using the function: $f(x) = 3x^2 + 1$